
Temporal and spatial variation of groundwater level in Mymensingh district, Bangladesh

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Abstract: In this paper, spatiotemporal analysis of groundwater level fluctuations of 32 piezometric wells using geostatistical analysis was done for Mymensingh district. A total nine years of weekly ground water level data were used for the analysis. Geostatistical analysis was performed using ordinary kriging and empirical Bayesian kriging (EBK) methods. The semivariogram models called spherical, exponential and Gaussian model were fitted with the experimental semivariogram in ordinary kriging while semivariogram fitting is automatic in EBK. Model performances were tested using root mean square standardised error (RMSSE), root mean square error (RMSE) and average standard error (ASE). The cross-validation results indicate that EBK performs better comparing to ordinary kriging in representing the spatial groundwater level fluctuation in the study area. The geostatistical analysis result shows that the Phulbaria, Trisal, Muktagachha, Bhaluka, Gafargaon and Mymensingh Sadar Upazila is comparatively more vulnerable than other parts of the district.

Keywords: geostatistical analysis; groundwater; ordinary kriging; empirical Bayesian kriging; EBK; Bangladesh.

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1 Introduction

Groundwater is one of the major sources of fresh water. One-third of world's total population depends primarily on groundwater for drinking, domestic, industrial, and agricultural purposes (Ahmadian and Chavoshian, 2012). The rate of extraction of groundwater is continuously increasing with the escalation of the population, agriculture and industrial development. This excessive groundwater extraction is responsible for serious groundwater level declination in many areas (Shamsudduha et al., 2009; Phien-wej et al., 2006; Bui et al., 2012; Akther et al., 2009). The declination of groundwater level has adverse effects on water pollution, salinity intrusion, groundwater contaminant, and imbalance of ecosystem, etc. (Baalousha, 2011; Antonellini et al., 2008; Bui et al., 2012). Therefore, a regular monitoring of groundwater levels is quite important to cope with the current and future water demand. However, remotely placed observation wells do not provide detailed information about spatial coverage of water level. Recently,

spatial interpolation mechanism through geostatistical analysis has become very popular to analyse such spatial characteristics of the groundwater level.

Geostatistical analysis has been widely used in hydrology to determine various important properties of the aquifer (Webster and Olive, 2007; Akhondi and Mohammadi, 2014). Besides, geostatistics-based spatiotemporal variability analysis is used successfully in many other fields (Taany et al., 2009; Machiwal et al., 2012; Ahmadi and Sedghamiz, 2007). Geostatistical analysis has been theoretically defined and applied by many researchers (Isaaks and Srivastava, 1989; Goovaerts, 1997; Kitanidis, 1997; Almedeij and Al-Ruwaih, 2006; Bui et al., 2012). Almedeij and Al-Ruwaih (2006) investigated the periodic behaviour of groundwater level fluctuations in the residential areas of Kuwait. Bui, et al. (2012) studied on spatiotemporal trend in the Red River Delta basin in Vietnam. They found that strongly declining groundwater level trends were mainly found in urban areas having intensive abstraction. Application of geostatistics in groundwater level analysis is found in Bangladesh also. Akther et al. (2009) identified the special and temporal variation of groundwater level in Dhaka city and found a severe declination in central Dhaka. Shamsudduha et al. (2009) analysed the spatial variation of trend and seasonal component and found a declining groundwater level trend (> 1 m/yr) in urban and peri-urban areas around Dhaka as well as in north-central, northwestern, and southwestern parts of the country.

Mymensingh is one of the most developing districts of Bangladesh. Population growth and industrialisation are increasing rapidly in this area. As a result, tremendous pressure on groundwater is being experienced in this region. While the groundwater level is found to be depleting over the past decades, no study has been done for this area addressing the temporal and spatial variation of groundwater. It is therefore aimed in this current study to investigate the spatiotemporal variation of groundwater level in Mymensingh district using geostatistical analysis.

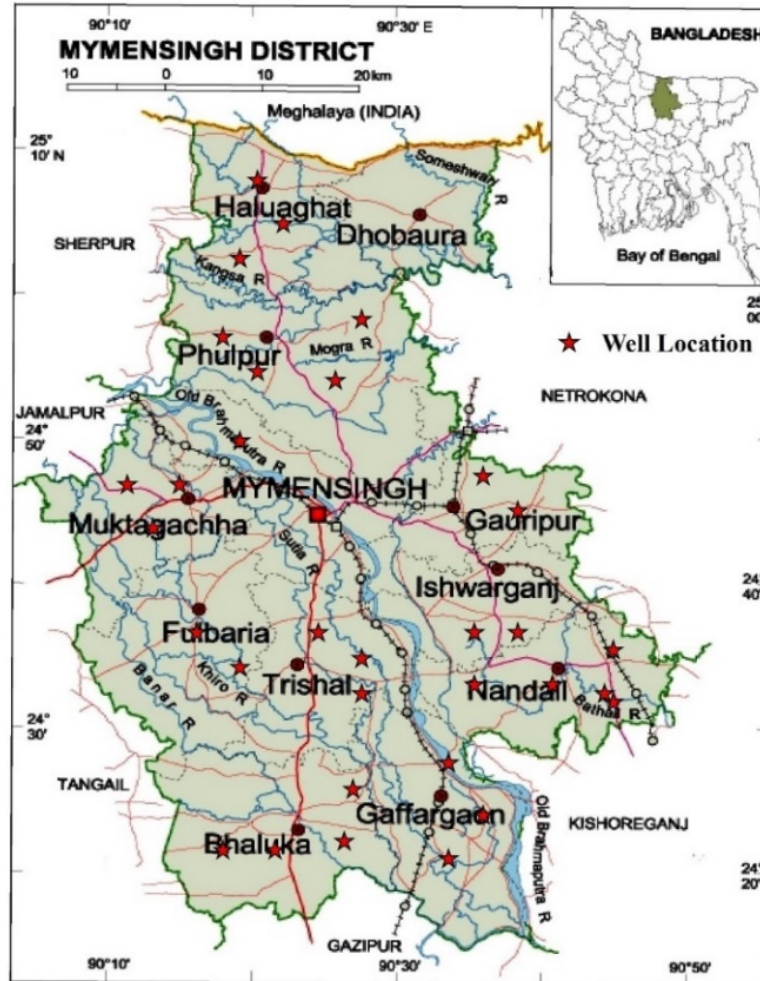
2 Study area and data collection

Mymensingh is located at $24^{\circ}15' - 25^{\circ}12'$ N, $90^{\circ}04' - 90^{\circ}49'$ E. The district covers an area of 4,396.53 km². It is bounded by Netrokona and Kishoregonj on the east, Sherpur, Jamalpur and Tangail on the west, Gazipur on the south and Garo hills of Meghalaya of India on the north. The city of Mymensingh stands on the bank of the Old Brahmaputra River. The temperature ranges from 18 to 34°C, and the annual rainfall averages 2,541 mm (BBS, 2013). The shallow aquifers in the Mymensingh region are characterised by approximately neutral pH, moderate COD (chemical oxygen demand), and high concentrations of arsenic (As) and iron (Fe). Arsenic concentrations in groundwater are highly variable (8–251 µg/l), and in many places exceed the maximum permissible limits of the WHO (10 µg/l) and Bangladesh (50 µg/l) for drinking water (Ahmed et al., 2010).

Weekly groundwater level data at 38 wells during 1999 to 2007 were collected from Bangladesh Water Development Board (BWDB). BWDB measures groundwater level at each station from the common horizontal datum, known as the Public Works Datum (PWD) (Shamsudduha et al., 2009). Collected data were then checked for missing values. A station series was considered complete if it has equalled to or greater than 75% observations available. Thirty-two well passed the above mentioned criterion, and these were used in this study. Weekly groundwater level data for these 32 wells were then

averaged to obtain annual mean ground water level. It will be worthy to mention here that averaging smooth the impact of missing values. As we have used annual average ground water level in the analysis, missing values do not have a substantial impact. Besides, the illustrative years in the geostatistical analysis were free from missing observations. The well locations in the study area are shown in Figure 1.

Figure 1 Study area with well locations (see online version for colours)



3 Methodology

The spatial prediction maps of water level were generated using kriging method. There are various kriging methods available (Isaaks and Srivastava, 1989). In this research, ordinary kriging and empirical Bayesian kriging (EBK) were used. Ordinary kriging is a widely used kriging method (Parvin et al., 2011). In contrast, EBK is a new type of kriging method that was first implemented in 2012 at ArcGIS 10.1 (Krivoruchko, 2012).

software package. EBK is a geostatistical interpolation method that automates the most difficult aspects of building a valid kriging model (Krivoruchko, 2012). The most desirable benefit of EBK is that the semivariogram models need not fit manually with the experimental semivariogram.

Whether it is ordinary Kriging or EBK, the data distribution must be normal for geostatistical analysis. Most common techniques for testing normality is to check histogram plot, normal Q-Q plot, Kurtosis, Skewness. In addition to these, more powerful Anderson-Darling test for normality was also used in the current analysis (Coronel-Brizio et al., 2010; Razali and Wah, 2011; Stephens, 1974).

The experimental semivariogram for ordinary kriging was fitted to inspect spatial dependency among measuring sites. This semivariogram is defined as the one-half of the average squared difference between the paired data values (Wackemagel, 2003; Mini et al., 2014). Mathematical expression of experimental semivariogram is as following:

$$\gamma(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [Z(x_i) - Z(x_i + h)]^2$$

where $\gamma(h)$ is the experimental semivariogram value at distance interval h ; $N(h)$ is the number of sample pairs within the distance interval h ; $Z(x_i)$, $Z(x_i + h)$ are the sample value at two points separated by the distance interval h .

The resulting semivariogram was then fitted with the most common standard models (e.g., spherical, exponential and Gaussian model). The standard model that best fitted with experimental semivariogram has been selected for ordinary kriging.

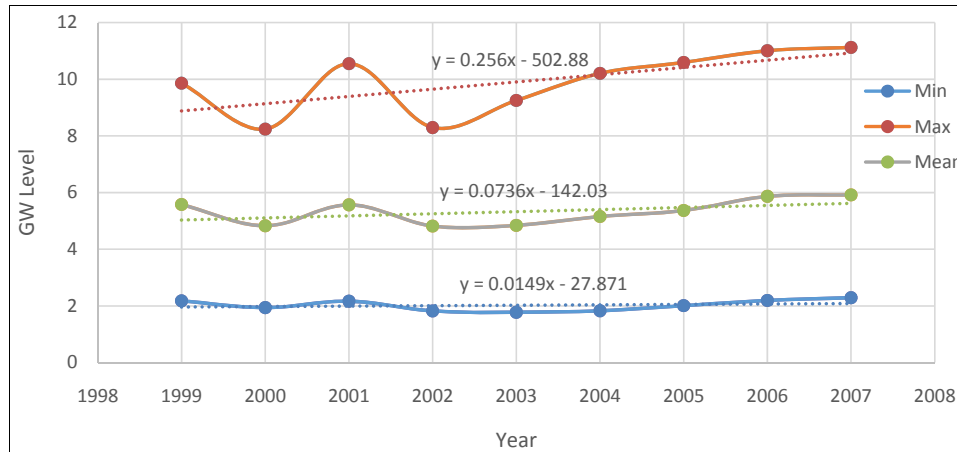
For comparison between various types of semivariograms used in ordinary kriging (Taany et al., 2009; Machiwal et al., 2012) and EBK, cross-validation criteria such as root mean square error (RMSE), root mean square standardised error (RMSSE) and average standard error (ASE) were used (Liao et al., 2006; Uyan and Cay, 2013).

In this study, descriptive statistics and normality test (Anderson-Darling test) were performed using MATLAB r2013b, Semivariogram were calculated by GS+ (Version 5.1.1), and kriging method were performed using ArcGIS 10.1.

4 Result and discussion

4.1 Water level trend

The annual minimum, maximum and mean values of ground water level over various time intervals are plotted in Figure 2. The linear regression trend line of minimum, maximum and mean values shows an increasing trend indicates that the groundwater level is lowering day by day. The slope of the linear trend line of maximum, mean and minimum ground water level are 0.256, 0.0736 and 0.0149 m/year, respectively.

Figure 2 Change of groundwater level with time (see online version for colours)

4.2 Normality of data

For the spatial analysis purpose, basic properties of water level data were calculated at first to assess normality of the series. Year wise descriptive statistics, including minimum, maximum, mean, median, standard deviation, skewness, and kurtosis are listed in Table 1. The table shows a very close mean and median value and a nominal skewness (near to 0) indicating any symmetrical distribution of water level data. Histograms and normal QQ Plot demonstrate that groundwater level fluctuation data approximately follow the normal distribution. Also, the Anderson-Darling test confirms normal distribution (Anderson-Darling test p-value > 0.05) (Theodorsson, 1988) of data.

4.3 Anisotropy test

To check the anisotropy in the groundwater level drop, variograms in several directions (0° , 45° , 90° , and 135° with a tolerance angle of $\pm 22.5^\circ$) were compared. No significant differences were found among the structures of the calculated variograms in all the four directions. Therefore, the spatial variation of groundwater level drop was considered to be isotropic while performing the geostatistical analysis.

4.4 Spatio-temporal variation in ground water level

The performance of three semivariogram models (spherical, exponential, Gaussian) intended to use for ordinary Kriging, and the one auto fitted in EBK (or simply EBK performance) were compared with each other. While comparing, RMSSE was chosen as the main cross-validation parameter. An RMSSE value closer to unity indicates a well-fitted semivariogram model. If $\text{RMSSE} < 1$, there is a tendency towards overestimation of the variance, if $\text{RMSSE} > 1$, there is a tendency toward underestimation (Liao et al., 2006). However, RMSE and ASE were also used while comparing performance. When RMSE and ASE are found equal or close enough, the model is considered as satisfactory (ESRI, 2012). The validation results of three

semivariogram models for ordinary kriging and EBK are listed in Table 2. According to the results shown in Figure 2, EBK gives the best result for the year 1999, 2000, 2002, 2004 and 2005. Gaussian model is found to be best for 2001 and 2007 while the Spherical model is found best for 2003 and 2006. Best results are shown in bold font in Table 2.

Table 1 Descriptive statistic of groundwater levels during 1999 to 2007.

<i>Year</i>	<i>Number of well</i>	<i>Min</i>	<i>Max</i>	<i>Mean</i>	<i>Median</i>
1999	32	2.1806	9.8625	5.5798	5.5737
2000	32	1.9406	8.2392	4.8268	4.7809
2001	32	2.1673	10.5529	5.5688	5.7226
2002	32	1.8212	8.2963	4.8179	4.9204
2003	32	1.7763	9.2533	4.8423	4.5819
2004	32	1.8285	10.2098	5.1551	5.1794
2005	32	2.0106	10.5952	5.3685	5.2876
2006	32	2.1942	11.0076	5.8677	5.8669
2007	32	2.2908	11.1268	5.9185	6.154
<i>Year</i>	<i>Standard deviation</i>	<i>Skewness</i>	<i>Kurtosis</i>	<i>Anderson-darling test p-value</i>	<i>Normally distributed (yes/no)</i>
1999	2.0625	0.14115	2.2188	0.4824	Yes
2000	1.7633	0.05999	2.1317	0.7656	Yes
2001	2.1802	0.28593	2.4759	0.8337	Yes
2002	1.8832	0.15241	1.982	0.3774	Yes
2003	1.968	0.3267	2.2878	0.317	Yes
2004	2.2056	0.36817	2.3379	0.6674	Yes
2005	2.2342	0.2691	2.327	0.7339	Yes
2006	2.3113	0.30914	2.2879	0.7539	Yes
2007	2.2651	0.10314	2.2152	0.2407	Yes

Figure 3 Spatial variability maps of groundwater level in 2000, 2003 and 2007 (see online version for colours)

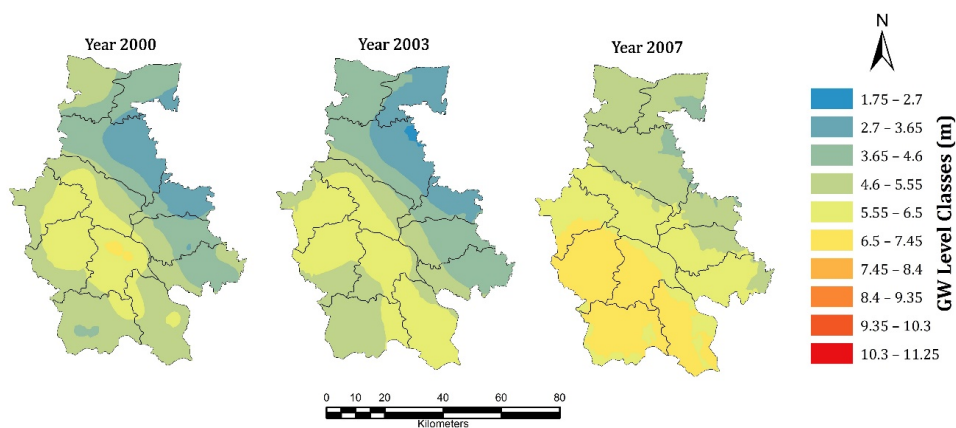


Table 2 Cross-validation results of ordinary kriging and EBK

Kriging method		Stat. index	Time								
			1999	2000	2001	2002	2003	2004	2005	2006	2007
OrdK	Sprc	RMSE	1.797	1.535	2.095	1.728	1.857	2.012	2.09	2.277	2.35
		RMSSE	0.951	0.962	0.955	0.953	0.996	0.945	0.972	1.004	0.992
		ASE	1.908	1.615	2.206	1.825	1.874	2.146	2.159	2.274	2.37
	Exp	RMSE	1.809	1.566	2.074	1.718	1.871	2.005	2.106	2.269	2.336
		RMSSE	0.942	0.931	0.949	0.912	0.954	0.952	0.952	0.967	0.962
		ASE	1.941	1.705	2.203	1.892	1.983	2.125	2.227	2.364	2.435
	Gaus	RMSE	1.777	1.52	2.082	1.698	1.844	2.004	2.079	2.264	2.345
		RMSSE	0.964	0.988	1.001	0.982	0.981	0.975	0.96	1.005	0.993
		ASE	1.858	1.557	2.084	1.738	1.886	2.067	2.173	2.257	2.363
EBK		RMSE	1.827	1.572	2.093	1.744	1.862	2.017	2.083	2.27	2.288
		RMSSE	0.984	0.989	0.983	0.998	0.986	0.982	0.986	0.989	0.991
		ASE	1.861	1.598	2.134	1.751	1.894	2.056	2.119	2.298	2.31
Best model			<i>EBK</i>	<i>EBK</i>	<i>Gaus</i>	<i>EBK</i>	<i>Sprc</i>	<i>EBK</i>	<i>EBK</i>	<i>Sprc</i>	<i>Gaus</i>

To compare the spatial and temporal groundwater level variation in the study area, three groundwater depth maps were exported for the year 2000, 2003 and 2007. They are shown in Figure 3. The figure indicates that the northeast region of Mymensingh district is facing low water level depletion than that of the southwest region (Phulbaria, Trisal, Muktagachha, Bhaluka, Gafargaon, and Mymensingh Sadar Upazila). Albeit, the water level is depleting in the whole area.

5 Conclusions

The geostatistical analysis is beneficial for sustainable water resources development, management, and supervisions. This paper demonstrates performance and application of newly established EBK for the spatiotemporal groundwater variability analysis in addition to existing methods. It is found that EBK has better accuracy in most of the cases. The paper also exhibits the spatiotemporal variation of ground water level in the study area. The result shows that groundwater level is lowering day by day in the study area. Linear trend analysis results indicate that maximum, mean and minimum ground water level are lowering at a rate of 0.256, 0.0736 and 0.0149 m/year. It also found that depletion of groundwater level is high in the south west region in Mymensingh district (Phulbaria, Trisal, Muktagachha, Bhaluka, Gafargaon, and Mymensingh Sadar Upazila) over the study period.

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